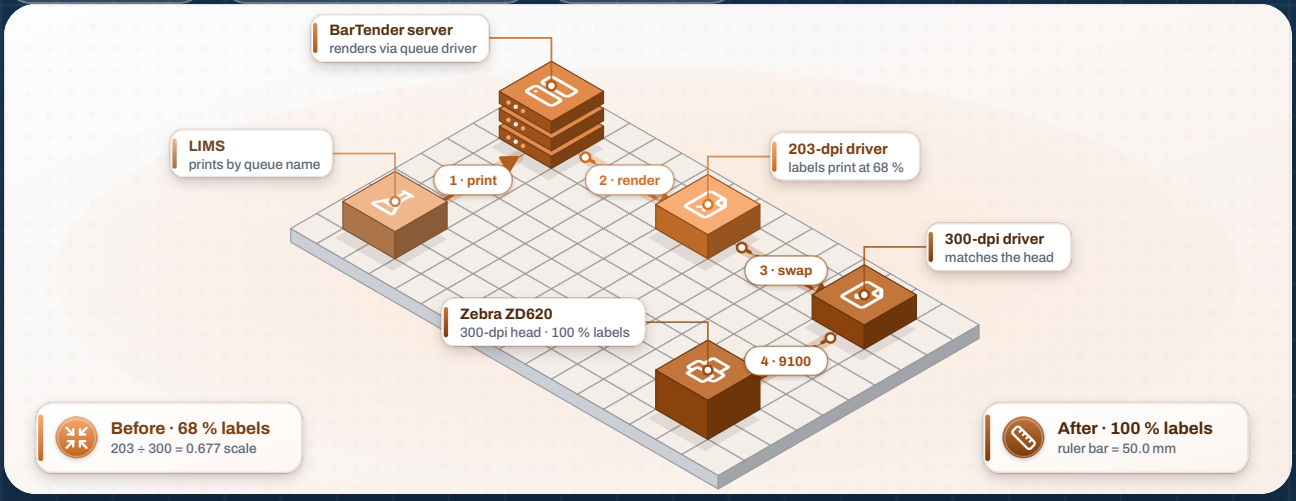
STANDARD OPERATING PROCEDURE · FIELD GUIDE SERIES

SOP-LBL-001

Fixing Undersized Labels: Zebra Printer Driver Resolution (DPI) Mismatch

Diagnose and correct a print-queue driver whose resolution does not match the printhead, four ways (Print Management GUI, a PowerShell base template, PrintUI/pnputil, and print-server standards with Group Policy), proven with a raw-ZPL 50 mm ruler test.

Label Printing | Windows Print Services | Lab Systems



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Sanitized reference edition — client identifiers removed

Contents

1. Concepts You Need First 3

1.1 Dots per inch: how a thermal printer draws 3

1.2 The journey of a label: LIMS to printhead 5

1.3 ZPL and EPL: the printer's languages 6

1.4 Seagull versus ZDesigner drivers 7

1.5 Why front-panel changes revert 7

1.6 Print queues, drivers and ports in Windows 8

1.7 Choosing a method 9

2. Access, Prerequisites and Diagnosis 11

2.1 Level of access required 11

2.2 Prerequisites checklist 12

2.3 Diagnosis: proving the mismatch 12

3. Method A · Print Management GUI 14

4. Method B · PowerShell Base Template 16

5. Methods C and D · Command Line and Prevention 19

5.1 Method C · PrintUI and pnputil 19

5.2 Method D · Print-server standards and Group Policy 19

6. Verification, Rollback and Troubleshooting 21

6.1 Test 1 · The 50 mm ruler test (proves the head) 21

6.2 Test 2 · End-to-end reprint (proves the queue) 22

6.3 Evidence for the change record 22

6.4 Rollback 22

6.5 Failover-queue parity 23

6.6 Prevention checklist 24

6.7 Troubleshooting 25

7. Summary, Glossary and Review 27

7.1 Chapter summary 27

7.2 Glossary 27

LEARNING OBJECTIVES

After working through this procedure you will be able to:

- Explain dots per inch (DPI), why a 203-dpi job printed on a 300-dpi head shrinks to 68 %, and how to spot the ratio in a defective label.
- Trace a label from the LIMS through BarTender, the Windows print queue and driver, TCP 9100 and the printer, and name where scale is decided.
- Distinguish ZPL from EPL and Seagull from ZDesigner drivers, and explain why front-panel changes revert on every job.
- Swap a queue to the correct driver through the GUI, a parameterized PowerShell template, and PrintUI/pnputil, with a backup first.
- Prove the printhead resolution independently of any driver with a raw ZPL ruler label sent over TCP 9100.
- Roll back safely and prevent recurrence with driver standards, failover-queue parity and Group Policy.

TIME	DIFFICULTY	MINIMUM ACCESS	RISK
≈ 30 min including tests	Intermediate	Local admin on the print / BarTender server	Low · queue name and port preserved

CHAPTER 1

Concepts You Need First

A thermal label printer looks simple: a roll of labels goes in, barcodes come out. Yet the most frustrating label faults are not mechanical at all. In the reference engagement a laboratory printer was moved to a new bench, re-registered in the Laboratory Information Management System (LIMS), and from then on every label came out at roughly two thirds of its intended size. The barcode no longer spanned the label, and text fields ran into each other ("Part Number100-2045"). Operators changed settings on the printer's front panel; the changes did nothing, and on the next label they had quietly reverted. Nothing was broken in the printer. The *Windows print driver* on the server believed it was talking to a different printer with a different resolution.

This SOP teaches you to recognise that fault in minutes, prove it, fix it four different ways and stop it coming back. First, the concepts.

1.1 Dots per inch: how a thermal printer draws

A direct-thermal or thermal-transfer printhead is a single row of tiny heating elements. Each element heats one **dot**; the label advances one dot row at a time and the printer heats the elements needed for that row. Everything on the label, from barcode bars to letters, is made of these dots. **Resolution** is how densely those dots are packed, quoted in **dots per inch (DPI)** or dots per millimetre.

Printhead	Dots per mm	Dot size (approx.)	Typical use
203 dpi	8	0.125 mm	Shipping labels, larger barcodes, general purpose
300 dpi	12	0.085 mm	Small specimen and part labels, 2-D codes, fine text
600 dpi	24	0.042 mm	Very small labels, electronics and medical-device marking

The same printer *model* is often sold in more than one resolution. A Zebra ZD620, for example, ships as a 203-dpi or a 300-dpi unit, and the two look identical on the bench. The resolution is printed on the configuration label and is visible in the printer's web page, but it is *not* obvious from the outside.

KEY CONCEPT · A PRINT JOB IS MEASURED IN DOTS, NOT MILLIMETRES

A label designer thinks in millimetres: "a 50 mm wide barcode". The printer only understands dots: "heat 590 elements". Something must translate millimetres into dots, and to do so it needs to know the dot density. On Windows that translator is the **printer driver**. If the driver believes the head has 203 dots per inch, it converts 50 mm into about 400 dots. A 300-dpi head then prints those 400 dots at its own, finer pitch, and the result measures only 400 ÷ 300 inches, about 34 mm. The label is not cropped; it is *shrunk*.



Figure 1. A driver set for fewer dots per inch than the head produces a job with fewer dots, so the same design prints uniformly smaller toward the top-left corner.

WORKED EXAMPLE · READING THE RATIO OFF A BAD LABEL

The defective labels in the reference case printed at roughly 60–68 % of the intended width. Compute the two possible mismatches:

- 203 ÷ 300 = **0.677**: a 203-dpi job on a 300-dpi head shrinks to 68 %.
- 300 ÷ 203 = **1.478**: a 300-dpi job on a 203-dpi head grows to 148 % and is cropped at the label edge.

A uniform shrink close to 0.68 is almost a fingerprint. Measure a reference element (barcode width or label border) and divide actual by expected before you touch anything.

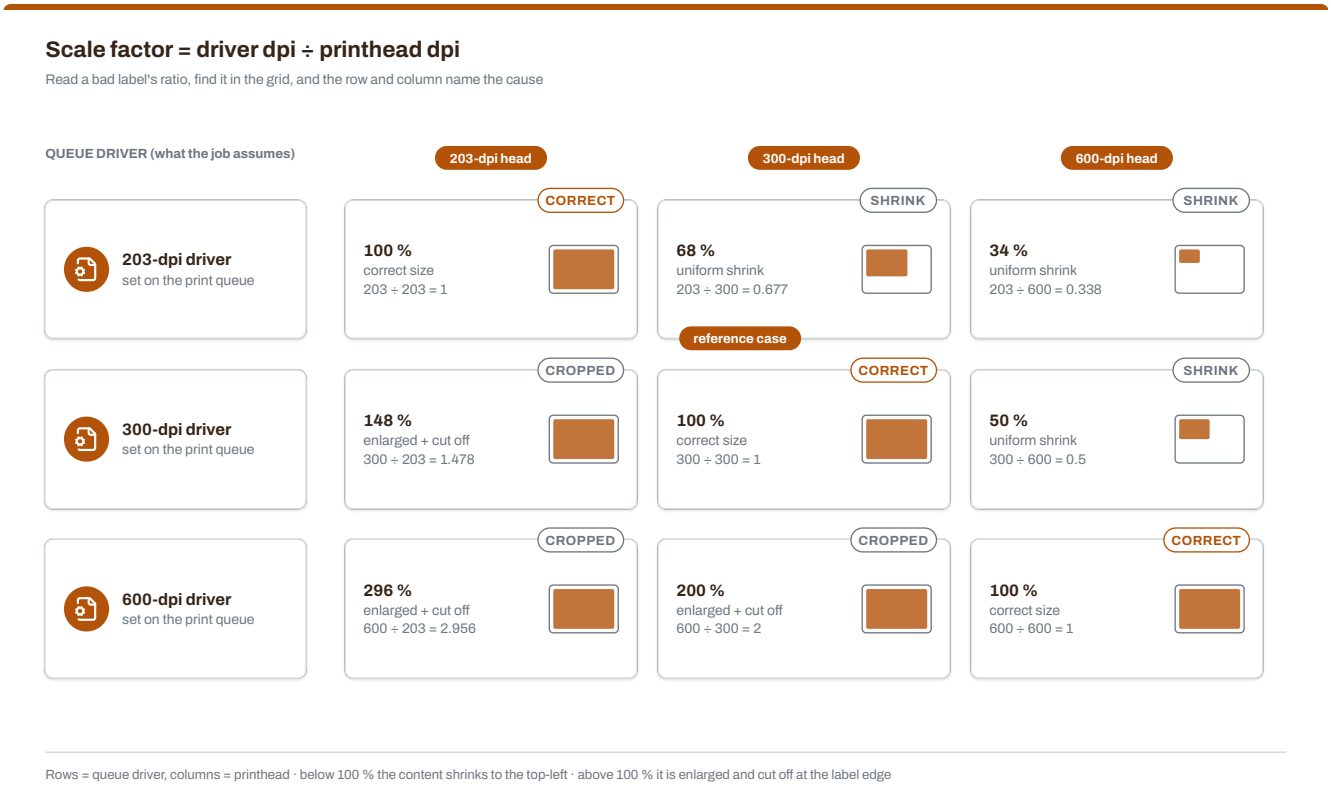


Figure 2. Divide the printed size by the designed size on a bad label and the matching cell names which driver and which head are paired.

1.2 The journey of a label: LIMS to printhead

In a regulated lab, users rarely print labels "directly". The LIMS (LabVantage in the reference environment) decides *what* to print, label-automation software (BarTender) decides *how it looks*, and Windows print services deliver the job to the device.

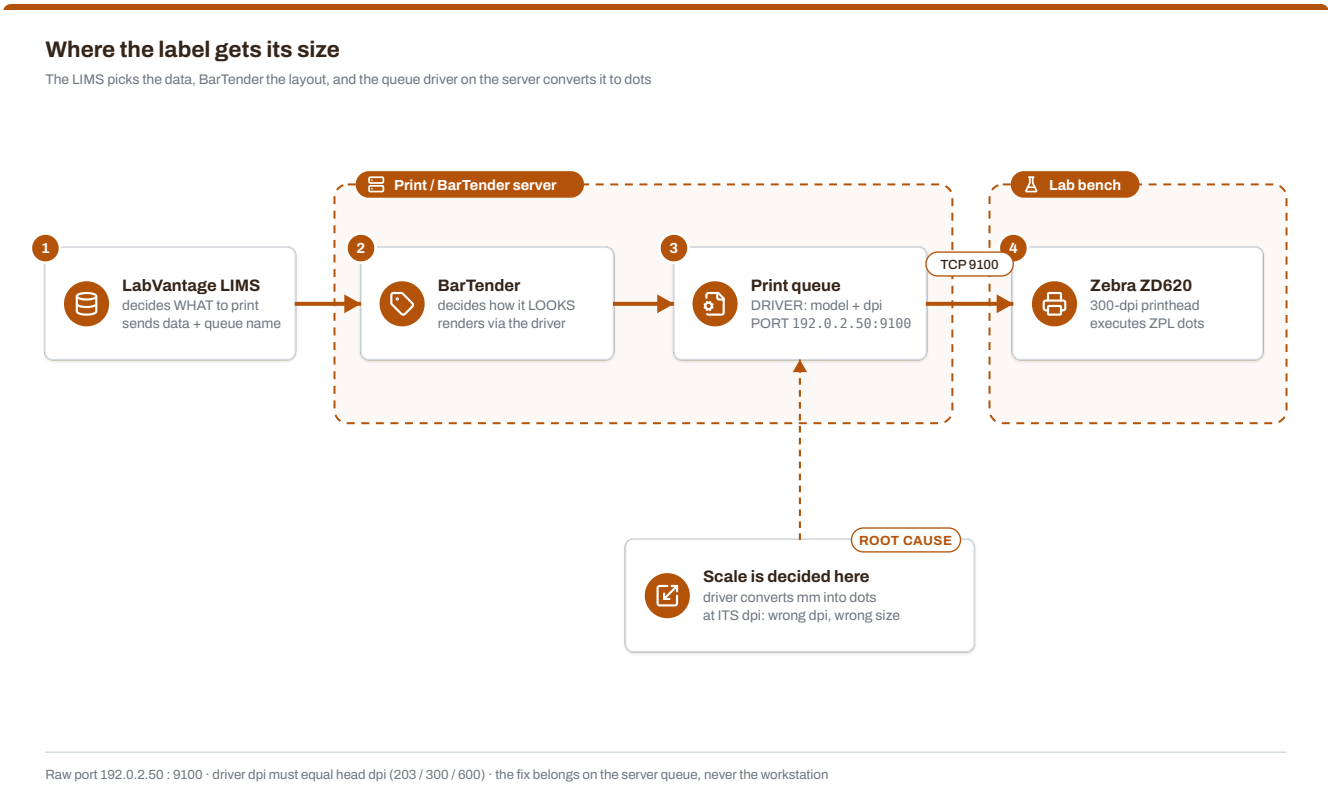


Figure 3. Only the queue driver on the print server turns millimetres into dots, so that driver is the one place a resolution mismatch can be fixed.

WHY IT MATTERS · THE FIX BELONGS ON THE SERVER

Because the driver on the BarTender server makes the dot conversion, the only place a resolution mismatch can be fixed is that driver. Nothing done at a workstation, in the LIMS, or on the printer's front panel changes how many dots the job contains.

1.3 ZPL and EPL: the printer's languages

Zebra printers speak a **page-description language**. The modern one is **ZPL** (Zebra Programming Language, also called ZPL II). Commands start with a caret, and a label sits between `^XA` (start) and `^XZ` (end). The older **EPL** (Eltron Programming Language) uses a different syntax and is retained on many models only for compatibility.

A MINIMAL ZPL LABEL, ANNOTATED — ZPL

```
^XA                        start of label format
^PW609                     print width = 609 dots (2.03 in on a 300-dpi head)
^LL300                     label length = 300 dots
^FO30,30^GB590,8,8^FS      field origin x=30,y=30; graphic box 590 dots wide, 8 high
^FO30,60^A0N,40,40^FDLINE = 50MM IF 300DPI^FS   scalable font 0, 40-dot text
^XZ                        end of label format
```

Note that every coordinate is in *dots*. Drivers come in ZPL and EPL variants for the same model. BarTender formats and the LIMS in this environment emit ZPL, so the **ZPL variant is mandatory**; an EPL driver produces jobs whose commands, fonts and default behaviour differ from every other printer on the site, even when the printer can interpret them.

1.4 Seagull versus ZDesigner drivers

Driver family	Who makes it	Strengths	Typical name format
Seagull ("Drivers by Seagull")	Seagull Scientific, the maker of BarTender	Tight BarTender integration (printer fonts, stock lists, BarTender-aware features). Preferred on BarTender servers.	`Zebra ZD620-300dpi` (exact text varies by release)
ZDesigner	Zebra Technologies	Vendor-supported, signed, used by Zebra utilities; fine for general Windows printing.	`ZDesigner ZD620-300dpi ZPL`

Both families embed the resolution in the model name. That is useful: **the driver name on the queue tells you what resolution the server believes**. In the reference case the queue was set to a *ZT620 (203 dpi)* model, while the device at the queue's IP address was a *ZD620* with a 300-dpi head: wrong model and wrong resolution. Do not mix families for the same model on one server; it makes audits and troubleshooting harder.

1.5 Why front-panel changes revert

Operators naturally try to fix size problems on the printer: print width, label length, darkness. On a BarTender-driven printer those changes appear to work and then vanish. The reason is **command precedence**.

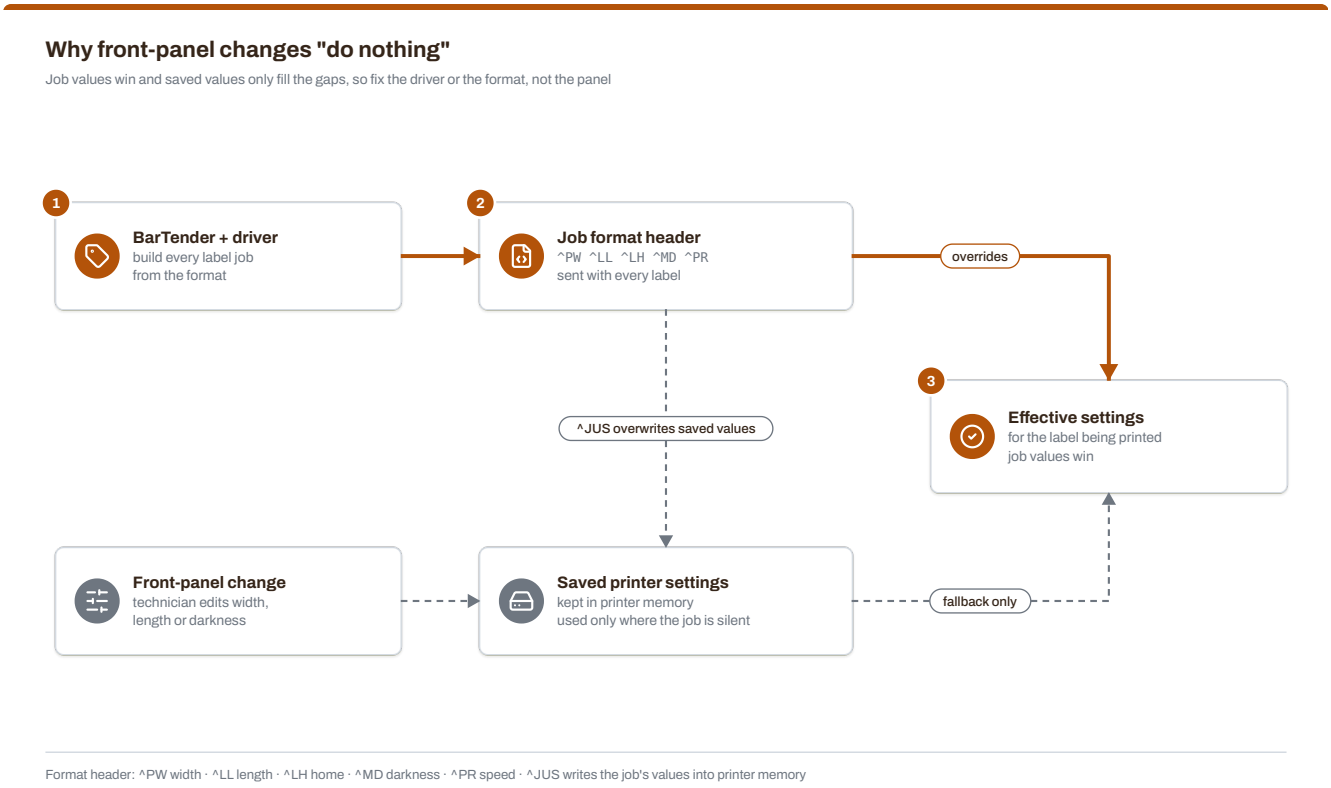


Figure 4. A setting that keeps reverting is being sent by every job, so change the driver or the BarTender format rather than the printer panel.

BarTender (through the driver) sends a full format header with every job: print width (`^PW`), label length (`^LL`), label home (`^LH`), darkness (`^MD`), speed (`^PR`) and more. Job commands override the values stored in the printer. Some drivers and formats also send `^JUS` ("save current settings"), which writes the job's values back into the printer's memory, so the front-panel change is literally overwritten. The lesson: *if a setting keeps reverting, the job is setting it, and the fix belongs where the job is built.*

1.6 Print queues, drivers and ports in Windows

A Windows print queue binds together three independent objects. The **driver** is installed once per server and referenced by any number of queues; the **port** (a Standard TCP/IP Port) holds the printer's IP address and raw port 9100; the **queue** carries the name the LIMS knows. Because they are independent, you can replace a queue's driver without touching its name, port or permissions, so no LIMS re-registration is needed.

WATCH OUT · HOW THE REFERENCE FAULT WAS BORN

During a lab relocation the queue was repointed (new port address) to a different physical unit, but the driver was left as it was. The queue name, port and LIMS registration were all correct, so every "connectivity" check passed. Only the driver model was wrong. Always re-check the driver when the device behind a queue changes.

1.7 Choosing a method

A · Print Management GUI

Install the driver with the Add Printer Driver wizard and select it on the queue's Advanced tab.

BEST FOR: ONE QUEUE, LEARNING

B · PowerShell template

Backs up, reads the head resolution over the network, installs and swaps the driver, writes evidence.

BEST FOR: REPEATABLE, AUDITED FIXES; MANY QUEUES

C · PrintUI + pnputil

Built-in command-line tools usable from CMD, batch files and Server Core.

BEST FOR: NO GUI, LEGACY SCRIPTS

D · Server standards + GPO

Driver standards, naming, audits and Group Policy so the mismatch cannot recur.

BEST FOR: PREVENTION ACROSS A SITE

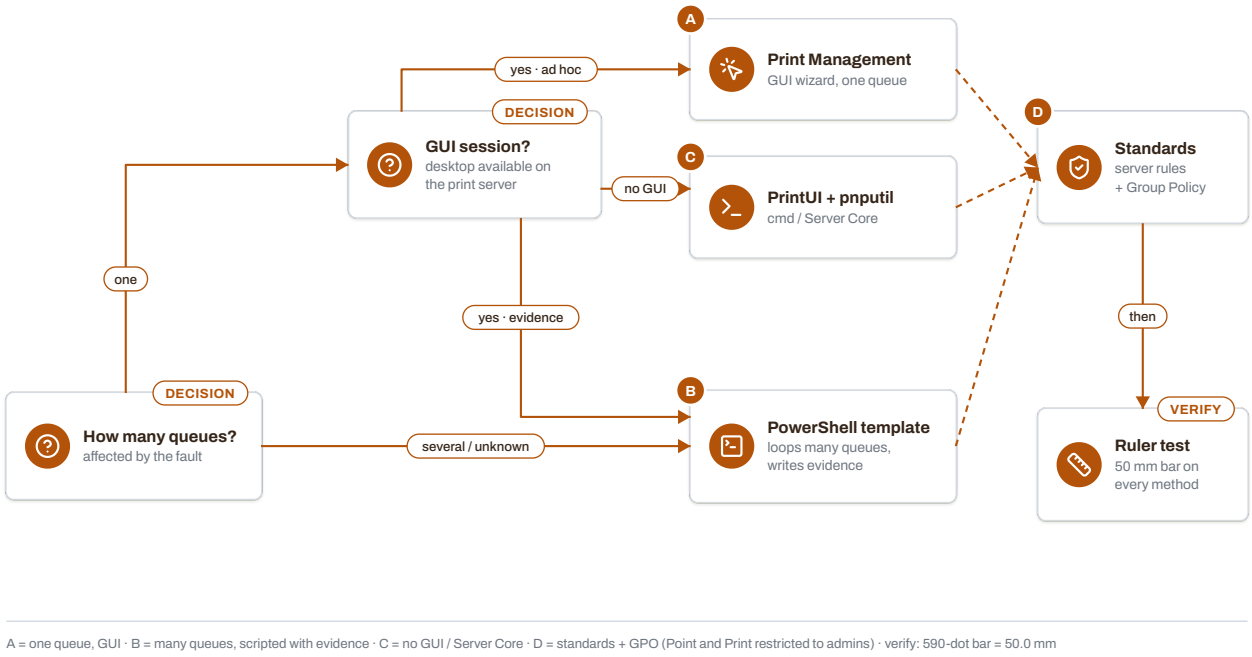


Figure 5. Pick the method by how many queues are affected and whether you have a desktop session, then add Method D so the fault cannot recur.

CHAPTER 2

Access, Prerequisites and Diagnosis

2.1 Level of access required

Since the 2021 "PrintNightmare" vulnerabilities, Windows restricts printer-driver installation to administrators by default (Point and Print `RestrictDriverInstallationToAdministrators` = 1). Plan for an administrator session on the print server, and keep every other right as narrow as possible.

Task	Least-privilege access	Notes
Stage and install a driver (pnputil, Add-PrinterDriver, wizard)	Member of local **Administrators** on the print / BarTender server	Required by the post-2021 driver-installation hardening. Use a separate admin account, not your daily account.
Change the driver and printing defaults on a queue	**Manage Printers** permission on the queue (driver must already be installed)	In practice the same admin session performs both steps.
Interactive session on the server	Remote Desktop Users on the server, through the site's privileged remote access path (for example BeyondTrust PRA or a jump host)	Many regulated sites require remote sessions to go through a recorded gateway.
Send raw ZPL to TCP 9100	Network reachability only (firewall path to the printer)	No authentication exists on port 9100; this is why printer VLANs should be segmented.
BarTender page setup and formats	BarTender Designer rights granted in BarTender security settings (Administration Console)	Only needed if stock dimensions must be re-saved in the format.
LIMS printer registration	LIMS application administrator (label application team)	Not needed when the queue name is preserved, which this SOP ensures.
Preventive Group Policy (Method D)	Edit rights on the target GPO, delegated on the server OU	Group Policy Management Console on an admin workstation.

GOOD PRACTICE · JUST-IN-TIME ADMINISTRATION

Check out the print-server admin credential from your privileged access management vault, or use a Windows LAPS-managed local admin password, for the length of the change only. Record the checkout against the change number; it becomes audit evidence that the right person held the right access for the right time.

2.2 Prerequisites checklist

- [ ] Approved change or incident record; note its number for the evidence file (for example `CHG0012345`).
- [ ] Maintenance window agreed with the lab: swapping the driver briefly interrupts jobs spooled to the queue.
- [ ] Correct driver package downloaded and extracted on the server, e.g. `C:\Drivers\ZD620\` (Seagull preferred on BarTender servers; ZDesigner acceptable). **ZPL variant only.**
- [ ] Network path from the server to the printer on TCP 9100 (for the raw ruler test).
- [ ] A person at the printer to collect and measure test labels, with a ruler or calipers, and a known-good label from a reference printer for comparison.
- [ ] The failover or redirect queue for this printer identified (if one exists), so parity can be checked afterwards.

2.3 Diagnosis: proving the mismatch

Diagnosis comes before remediation. The goal is to turn "labels look small" into a measured, evidenced hypothesis you could defend in a change review. The steps below follow the order used in the reference engagement.

1. **Characterise the symptom.** Collect several defective labels and photographs. Check that the shrink is *uniform* (everything smaller by the same factor) and *repeatable* (identical on every label). Drifting position or random cropping points to media calibration instead.
2. **Compare with a reference.** Print the same label method on a known-good printer with the same stock. If the reference is correct, the format and the LIMS are exonerated and the fault is isolated to the affected queue or device.
3. **Measure the ratio.** Measure a reference feature (barcode width, a border) on the bad label and on the good one. A ratio near 0.68 or 1.48 is the resolution fingerprint described in Chapter 1.
4. **Record what the server believes.** On the BarTender server, read the queue's driver and port:

WHAT THE SERVER BELIEVES POWERSHELL

```
Get-Printer -Name 'LAB-TORSION-04 (small)' | Format-List Name, DriverName, PortName, PrinterStatus
Get-PrinterPort -Name (Get-Printer -Name 'LAB-TORSION-04 (small)').PortName |
  Format-List Name, PrinterHostAddress, PortNumber, SNMPEnabled
```

5. **Record what the printer is.** Identify the physical unit at that IP address (rating label, asset label, photographs). A model mismatch between the driver name and the device, such as ZT620 versus ZD620, is the first red flag.

6. **Confirm the head resolution.** Use any of these three independent sources:
   - *Configuration label*: on a ZD620, press and hold FEED and CANCEL for two seconds. The report shows the resolution as dots per millimetre: 8 = 203 dpi, 12 = 300 dpi.
   - *Web page*: browse to `http://<printer-IP>`; the printer configuration page lists the resolution.
   - *Host identification query*: send `~HI` to port 9100 and read the reply (script below).

7. **Explain the reverting settings.** If operators report that front-panel changes do not stick, note it as supporting evidence: the job carries its own format settings, so the fix must be on the server.

GET-ZEBRAHEADDPI.PS1 · ASK THE PRINTER FOR ITS RESOLUTION — POWERSHELL

```powershell
function Get-ZebraHeadDpi {
  param([Parameter(Mandatory)][string]$Address, [int]$Port = 9100)
  $client = [System.Net.Sockets.TcpClient]::new($Address, $Port)
  try {
    $client.ReceiveTimeout = 3000
    $stream = $client.GetStream()
    $cmd = [Text.Encoding]::ASCII.GetBytes('~HI')           # host identification
    $stream.Write($cmd, 0, $cmd.Length)
    Start-Sleep -Milliseconds 600
    $buf = New-Object byte[] 512
    $n = $stream.Read($buf, 0, $buf.Length)
    $reply = [Text.Encoding]::ASCII.GetString($buf, 0, $n).Trim([char]2, [char]3, ' ', "`r", "`n")
    # Reply format: model,firmware,dots-per-mm,memory  e.g.  ZD620-300dpi,V84.20.10Z,12,8176KB
    $dpm = [int]($reply.Split(',')[2])
    [pscustomobject]@{ Address = $Address; Reply = $reply; DotsPerMm = $dpm
                       HeadDpi = @{ 6 = 152; 8 = 203; 12 = 300; 24 = 600 }[$dpm] }
  } finally { $client.Close() }
}
Get-ZebraHeadDpi -Address 192.0.2.50
```

NOTE · READING THE REPLY

The `~HI` reply is wrapped in STX/ETX control characters, which the script trims. The exact model string varies by firmware, but the third field is consistently the dots-per-millimetre value. If the printer does not answer (some units are configured to ignore host queries, or port 9100 is filtered), fall back to the configuration label.

CHAPTER 3

Method A · Print Management GUI

The graphical method uses the Print Management console (`printmanagement.msc`) and the queue's own properties dialog on the BarTender server. It is the best way to learn the moving parts, because you see the driver, port and defaults side by side.

STEP 1 Back up the print server configuration

1. Open **Print Management** (from Win+R, `printmanagement.msc`) with an elevated admin session.
2. Right-click the server under **Print Servers** and choose **Export printers to a file…**. Save the `.printerExport` file to the change folder. This captures queues, drivers and ports, and can be re-imported with **Import printers from a file…**.
3. Open the affected queue's **Properties** and take a screenshot of the **Advanced** and **Ports** tabs as before-state evidence.

STEP 2 Install the correct driver

1. In Print Management expand the server, right-click **Drivers** and choose **Add Driver…**. The Add Printer Driver Wizard opens.
2. Select the processor architecture (*x64*), then **Have Disk…** and browse to the extracted package, e.g. `C:\Drivers\ZD620\`, selecting the `.inf` file.
3. From the model list choose the model **and** resolution that match the device, in the **ZPL** variant: for example *ZD620-300dpi* (ZPL). Never choose an EPL entry.
4. Finish the wizard and confirm the new driver now appears in the **Drivers** list with its version and provider.

Vendor installers (Seagull Driver Wizard, ZDesigner installer) are valid alternatives to Have Disk; run them in "install driver only" mode so they do not create a duplicate queue.

STEP 3 Swap the driver and reset printing defaults

1. Under **Printers**, right-click the queue (e.g. `LAB-TORSION-04 (small)`) › **Properties** › **Advanced** tab.

2. In the **Driver** drop-down, select the newly installed 300-dpi ZPL driver. Select **Apply**. Windows may warn that the new driver's defaults replace the old ones; accept.
3. On the **Ports** tab, confirm the same TCP/IP port is still ticked and points at the printer's address. On the **General** tab, confirm the queue name is unchanged.

Reset the printing defaults

A newly assigned driver starts at *factory defaults*. Scale will be correct, but the stock size, orientation or darkness may not be, which can shift the label even though it is now the right size.

1. **Advanced › Printing Defaults...** (defaults for every user and for BarTender's service account; preferences under *Printing Preferences* only affect your own profile).
2. Set the stock or page size to the physical label (e.g. 2.0 × 1.0 in), the orientation, media type (direct thermal or thermal transfer) and darkness to match the reference printer.
3. Apply, then print one Windows test page from the **General** tab only if the lab agrees to waste a label; otherwise go straight to Chapter 6.

WATCH OUT · DO NOT "FIX" SCALE WITH A SCALING OPTION

Some drivers and BarTender page setup offer scaling or "fit to page". Using them to compensate for a wrong driver produces labels that look right but have barcodes with the wrong module width, which can fail verification or scanning. Fix the resolution, never the symptom.

CHAPTER 4

Method B · PowerShell Base Template

The template below is a generalized, hardened version of the script used in the reference engagement. It works for any Zebra model and resolution, supports `-WhatIf`, refuses EPL drivers, reads the head resolution from the printer itself, and writes a JSON evidence file for the change record. It uses only the built-in **PrintManagement** module and `pnputil.exe`, so nothing needs to be installed.

REPAIR-ZEBRAQUEUEDPI.PS1 · BASE TEMPLATE POWERSHELL

```powershell
<#
.SYNOPSIS  Swap a Zebra print queue to the driver that matches its printhead resolution (DPI).
.DESCRIPTION
  Stage 0  Read the queue and port; ask the printer for its real resolution (~HI on TCP 9100).
  Stage 1  Back up the queue (properties, print configuration, PrintUI settings file).
  Stage 2  Find the matching ZPL driver in the driver store, or stage it from an INF package.
  Stage 3  Swap the driver on the queue (name, port, share and permissions are preserved).
  Stage 4  Optionally print a 50 mm raw-ZPL ruler label; write JSON evidence.
.NOTES     Run elevated ON THE PRINT / BARTENDER SERVER that hosts the queue.
           Access: local Administrators (driver install) + Manage Printers on the queue.
#>
[CmdletBinding(SupportsShouldProcess)]
param(
  [Parameter(Mandatory)] [string] $PrinterName,                # e.g. 'LAB-TORSION-04 (small)'
  [Parameter(Mandatory)] [string] $ExpectedAddress,            # e.g. '192.0.2.50'
  [string] $ModelPattern    = 'ZD620',                         # model text as it appears in driver
names
  [ValidateSet(0,203,300,600)] [int] $HeadDpi = 0,             # 0 = ask the printer (~HI)
  [string] $DriverInfFolder = 'C:\Drivers\ZD620',              # extracted Seagull or ZDesigner package
  [string] $ChangeRef       = 'CHG0000000',
  [string] $EvidenceFolder  = 'C:\Temp\PrinterChanges',
  [int]    $RawPort         = 9100,
  [switch] $SendRulerTest
)
$ErrorActionPreference = 'Stop'
$stamp = Get-Date -Format 'yyyyMMdd_HHmmss'
New-Item -ItemType Directory -Path $EvidenceFolder -Force | Out-Null

function Send-Raw([string]$Address, [int]$Port, [string]$Data, [switch]$Read) {
  $c = [System.Net.Sockets.TcpClient]::new($Address, $Port)
  try {
    $c.ReceiveTimeout = 3000; $s = $c.GetStream()
    $b = [Text.Encoding]::ASCII.GetBytes($Data); $s.Write($b, 0, $b.Length)
    if ($Read) { Start-Sleep -Milliseconds 600; $buf = New-Object byte[] 512
                 $n = $s.Read($buf, 0, 512); [Text.Encoding]::ASCII.GetString($buf, 0, $n) }
  } finally { $c.Close() }
}

# ---- Stage 0: inspect queue, port and printer -----------------------------
$printer = Get-Printer -Name $PrinterName
$port    = Get-PrinterPort -Name $printer.PortName
$address = if ($port.PrinterHostAddress) { $port.PrinterHostAddress } else { $printer.PortName }
if ($address -ne $ExpectedAddress) {
  Write-Warning "Port resolves to '$address', expected '$ExpectedAddress'. Confirm the physical printer
before continuing."
}
```

```
if ($HeadDpi -eq 0) {
  $reply = (Send-Raw $ExpectedAddress $RawPort '~HI' -Read).Trim([char]2, [char]3, ' ', "`r", "`n")
  $HeadDpi = @{ 8 = 203; 12 = 300; 24 = 600 }[[int]$reply.Split(',')[2]]
  if (-not $HeadDpi) { throw "Could not read resolution from '$reply'. Pass -HeadDpi from the
configuration label." }
  Write-Host "Printer reports: $reply  => $HeadDpi dpi"
}
Write-Host "Queue '$PrinterName' uses '$($printer.DriverName)' on $address; head is $HeadDpi dpi."

# ---- Stage 1: back up ------------------------------------------------------
$backup = Join-Path $EvidenceFolder "$ChangeRef-$stamp"
$printer | Select-Object * | ConvertTo-Json -Depth 3 | Out-File "$backup-queue.json"
Get-PrintConfiguration -PrinterName $PrinterName | Select-Object * | ConvertTo-Json | Out-File "$backup-
config.json"
Start-Process rundll32.exe -Wait -ArgumentList "printui.dll,PrintUIEntry /Ss /n `"$PrinterName`" /a
`"$backup-printui.dat`""

# ---- Stage 2: locate or stage the matching ZPL driver ------------------------
$isTarget = { param($n) $n -match $ModelPattern -and $n -match "$HeadDpi\s*-?\s*dpi" -and $n -notmatch
'EPL' }
$target = Get-PrinterDriver | Where-Object { & $isTarget $_.Name } | Select-Object -First 1
if (-not $target) {
  $infs = Get-ChildItem -Path $DriverInfFolder -Filter *.inf -Recurse
  if (-not $infs) { throw "No .inf files under '$DriverInfFolder'. Extract the driver package there
first." }
  # Find the exact model string inside the INF files (prefer an explicit ZPL entry)
  $models = Select-String -Path $infs.FullName -Pattern ('"([^"]*' + $ModelPattern + '[^"]*)"') -
AllMatches |
            ForEach-Object { $_.Matches } | ForEach-Object { $_.Groups[1].Value } | Sort-Object -Unique
  $model  = ($models | Where-Object { & $isTarget $_ } | Sort-Object { $_ -notmatch 'ZPL' } | Select-
Object -First 1)
  if (-not $model) { $models | ForEach-Object { Write-Host "  found: $_" }
                     throw "No $HeadDpi-dpi ZPL model for '$ModelPattern' in the package." }
  if ($PSCmdlet.ShouldProcess($model, 'Stage INF with pnputil and Add-PrinterDriver')) {
    foreach ($inf in $infs) {
      pnputil.exe /add-driver "$($inf.FullName)" /install | Out-Null
      if ($LASTEXITCODE -notin 0, 259, 3010) { throw "pnputil failed on $($inf.Name) (exit
$LASTEXITCODE)." }
    }
    Add-PrinterDriver -Name $model
    $target = Get-PrinterDriver -Name $model
  }
}
Write-Host "Target driver: $($target.Name)" -ForegroundColor Green

# ---- Stage 3: swap the driver on the queue -----------------------------------
$before = $printer.DriverName
if ($before -eq $target.Name) { Write-Host 'Queue already uses the target driver.' }
elseif ($PSCmdlet.ShouldProcess($PrinterName, "Set-Printer -DriverName '$($target.Name)'")) {
  Set-Printer -Name $PrinterName -DriverName $target.Name
}
$after = Get-Printer -Name $PrinterName

# ---- Stage 4: ruler test and evidence ----------------------------------------
if ($SendRulerTest -and $PSCmdlet.ShouldProcess($ExpectedAddress, 'Send 50 mm ZPL ruler label')) {
  # 590 dots = 50 mm on a 300-dpi head (73.8 mm on 203 dpi). Bypasses the driver entirely.
  Send-Raw $ExpectedAddress $RawPort '^XA^FO30,30^GB590,8,8^FS^FO30,60^A0N,40,40^FDLINE = 50MM IF
300DPI^FS^XZ'
}
[pscustomobject]@{
  ChangeRef = $ChangeRef; Operator = "$env:USERDOMAIN\$env:USERNAME"; Server = $env:COMPUTERNAME
  TimeUtc = (Get-Date).ToUniversalTime().ToString('o'); Queue = $PrinterName; Port = $printer.PortName
```

```
  Address = $address; HeadDpi = $HeadDpi; DriverBefore = $before; DriverAfter = $after.DriverName
  Status = "$($after.PrinterStatus)"; RulerTestSent = [bool]$SendRulerTest; BackupPrefix = $backup
} | ConvertTo-Json | Out-File "$backup-evidence.json"
Write-Host "Done. Reset Printing Defaults (stock size) and reprint from the LIMS." -ForegroundColor
Yellow
```

EXAMPLE · RUNNING THE TEMPLATE

INVOCATION (ELEVATED, ON THE BARTENDER SERVER) POWERSHELL

```powershell
.\Repair-ZebraQueueDpi.ps1 -PrinterName 'LAB-TORSION-04 (small)' -ExpectedAddress '192.0.2.50' `
   -ModelPattern 'ZD620' -DriverInfFolder 'C:\Drivers\ZD620' -ChangeRef 'CHG0012345' -WhatIf

# Review the preview, then run for real and print the ruler label:
.\Repair-ZebraQueueDpi.ps1 -PrinterName 'LAB-TORSION-04 (small)' -ExpectedAddress '192.0.2.50' `
   -ModelPattern 'ZD620' -ChangeRef 'CHG0012345' -SendRulerTest
```

The first run reads the printer and shows what would change without changing anything. The second run swaps the driver and prints one ruler label. The evidence JSON lands in `C:\Temp\PrinterChanges\`.

KEY CONCEPT · WHY `SET-PRINTER -DRIVERNAME` IS SAFE FOR THE LIMS

`Set-Printer` replaces only the driver reference on the existing queue object. The queue name, share name, port, security descriptor and location are untouched, so the LIMS and BarTender keep printing to the same name. Deleting and re-creating the queue would also fix the resolution, but it would change the queue's identity and could break registrations and permissions.

CHAPTER 5

Methods C and D · Command Line and Prevention

5.1 Method C · PrintUI and pnputil

Windows ships two command-line tools that pre-date the PowerShell print cmdlets and still work everywhere: `pnputil.exe` manages the driver store, and `printui.dll` (invoked through `rundll32`) exposes almost every function of the printer properties dialog. They are ideal in batch files, in remote CMD sessions and on Server Core.

SWAP-ZEBRADRIVER.CMD · ELEVATED COMMAND PROMPT — CMD

```
:: 1. Back up the queue's settings to a file you can restore later
rundll32 printui.dll,PrintUIEntry /Ss /n "LAB-TORSION-04 (small)" /a "C:\Temp\CHG0012345-queue.dat"

:: 2. Stage the driver package in the driver store, then list printer drivers
pnputil /add-driver "C:\Drivers\ZD620\*.inf" /subdirs /install
pnputil /enum-drivers /class Printer

:: 3. Install the printer driver by its exact model name (copy it from the INF)
rundll32 printui.dll,PrintUIEntry /ia /m "ZDesigner ZD620-300dpi ZPL" /f "C:\Drivers\ZD620\ZDesigner.inf"

:: 4. Point the queue at the new driver, then read the settings back to confirm
rundll32 printui.dll,PrintUIEntry /Xs /n "LAB-TORSION-04 (small)" DriverName "ZDesigner ZD620-300dpi ZPL"
rundll32 printui.dll,PrintUIEntry /Xg /n "LAB-TORSION-04 (small)" /f "C:\Temp\after.txt" & type
C:\Temp\after.txt

:: 5. Open Printing Defaults to reset the stock size
rundll32 printui.dll,PrintUIEntry /e /n "LAB-TORSION-04 (small)"
```

NOTE · NAMES MUST MATCH EXACTLY

The model string passed to `/m` and to `DriverName` must match the INF character for character, including spaces and hyphens. The INF file name shown (`ZDesigner.inf`) is illustrative; use the file in your package. `rundll32` reports most failures in a pop-up, so on Server Core or in unattended runs confirm the result with `/Xg` or `Get-Printer`. To see every PrintUI switch, run `rundll32 printui.dll,PrintUIEntry /?`.

TIP · WHOLE-SERVER BACKUP WITH PRINTBRM

`%WINDIR%\System32\spool\tools\PrintBrm.exe -B -F C:\Temp\server.printerExport` backs up every queue, driver and port on the server (the same engine as Print Management's export). `PrintBrm -R -F <file>` restores it. Take one before any bulk driver change.

5.2 Method D · Print-server standards and Group Policy

Methods A to C fix one queue. Method D removes the conditions that produced the fault: a driver chosen by guesswork during a relocation, nothing recording the head resolution, and

no check at handover. It combines server-side standards with Group Policy.

Server-side standards

Standard	How to implement
One driver family per model	On BarTender servers use Seagull drivers for every Zebra model; remove unused families from the driver store once no queue references them (Print Management › Drivers › Remove Driver Package).
Resolution in the record	Write the head resolution in the queue's *Location* or *Comment* field and on the asset label: `Set-Printer -Name '…' -Comment 'ZD620 · 300 dpi · 2x1 in stock'`.
Visible mismatches	In Print Management create a Custom Printer Filter (right-click *Custom Filters* › Add New Printer Filter) with criteria on *Driver Name*, e.g. "contains 203", to list every 203-dpi queue for review.
Scheduled audit	A weekly scheduled task loops over `Get-Printer`, extracts the dpi from each Zebra driver name, calls `Get-ZebraHeadDpi` (Chapter 2) against the port address and exports any mismatches to CSV for the service desk.
Handover check	No queue is released to the lab after a move, swap or new install until the 50 mm ruler test passes and the configuration label is filed with the change.

Group Policy settings

Group Policy cannot choose a driver for a queue, but it governs who can install drivers, which servers clients may trust and how shared queues reach workstations.

1. **Keep driver installation admin-only.** In Computer Configuration › Administrative Templates › Printers, set Limits print driver installation to Administrators to *Enabled*. This blocks well-meaning users from adding a "close enough" driver.
2. **Restrict Point and Print to approved servers.** Configure Point and Print Restrictions and Package Point and print - Approved servers with the fully qualified names of your print servers (e.g. `srv-print-01.contoso.com`).
3. **Deploy shared queues centrally.** Use User or Computer Configuration › Preferences › Control Panel Settings › Printers (Group Policy Preferences) to map the lab's shared queues, or right-click a queue in Print Management and choose Deploy with Group Policy... . Workstations then always use the server's driver, so a fix on the server applies everywhere at once.
4. **Confirm the policy applies** on a test workstation with `gpresult /h C:\Temp\gp.html`.

CHAPTER 6

Verification, Rollback and Troubleshooting

6.1 Test 1 · The 50 mm ruler test (proves the head)

This test bypasses BarTender and the Windows driver altogether. It sends a ZPL bar exactly 590 dots long straight to the printer. Because no driver is involved, the printed length depends only on the printhead's real resolution.

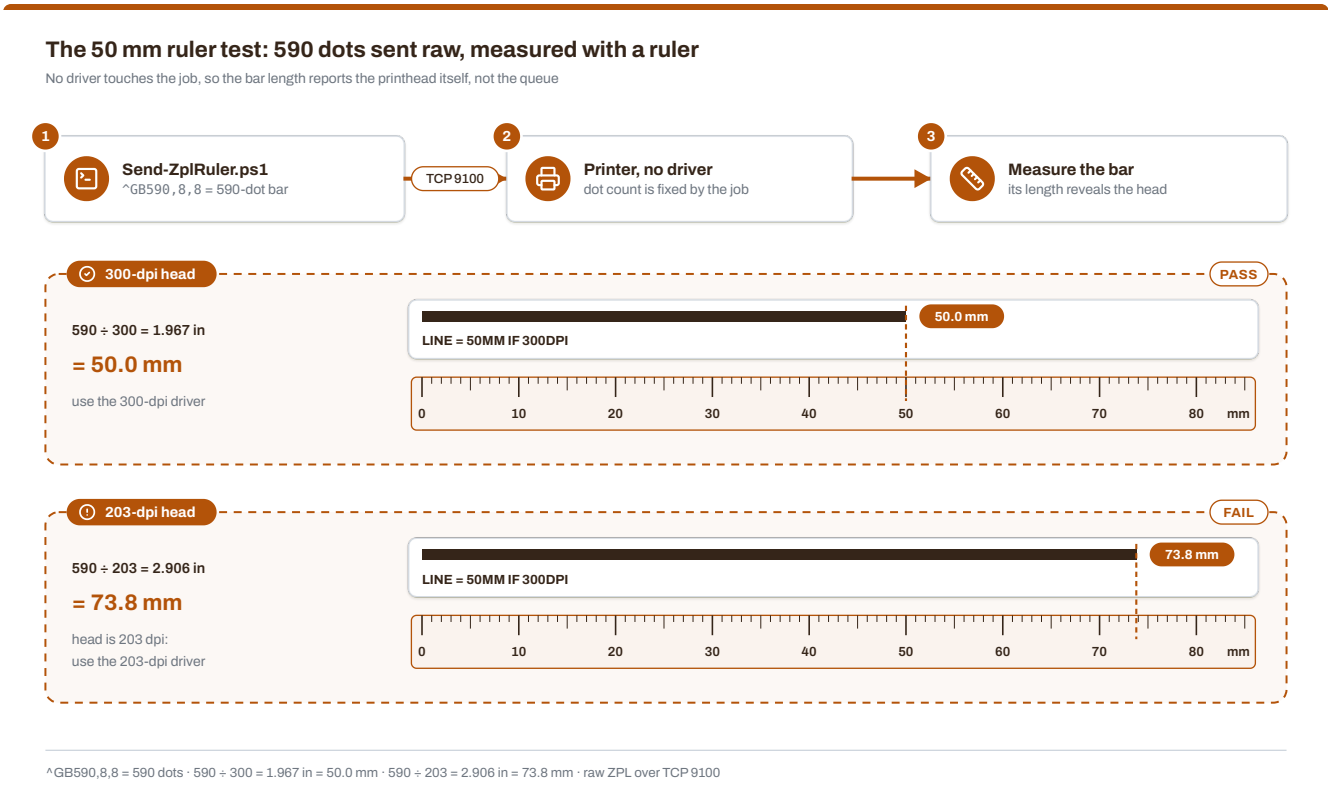


Figure 6. Because raw ZPL bypasses every driver, the measured bar length identifies the printhead resolution on its own.

SEND-ZPLRULER.PS1 · RAW TCP 9100 POWERSHELL

```powershell
$address = '192.0.2.50'
$zpl = '^XA^FO30,30^GB590,8,8^FS^FO30,60^A0N,40,40^FDLINE = 50MM IF 300DPI^FS^XZ'
$client = [System.Net.Sockets.TcpClient]::new($address, 9100)
$stream = $client.GetStream()
$bytes  = [System.Text.Encoding]::ASCII.GetBytes($zpl)
$stream.Write($bytes, 0, $bytes.Length)
$stream.Close(); $client.Close()
Write-Host 'Sent. Measure the bar: 50 mm = 300 dpi, ~74 mm = 203 dpi.'
```

Pick the bar length for other heads with *dots = mm × dpi ÷ 25.4*: 50 mm needs 394 dots on 203 dpi and 1181 dots on 600 dpi. If the label is narrower than the bar, shorten the target length (a 40 mm bar at 300 dpi is 472 dots).

NOTE · WHAT THE RULER TEST DOES AND DOES NOT PROVE

It proves the *head* resolution, which tells you which driver is correct. It does not prove the queue is fixed. Test 2 proves the queue.

6.2 Test 2 · End-to-end reprint (proves the queue)

1. Ask the lab to reprint the affected label methods from the LIMS (the same methods that failed, e.g. specimen and test-sample labels).
2. Check: barcode spans the intended width, text fields do not overlap, the label is positioned within the stock, and the barcode scans with the lab's handheld scanner.
3. Lay the new label over the known-good reference label; the outlines should match.

Finally confirm the queue state:

FINAL STATE POWERSHELL

```
Get-Printer -Name 'LAB-TORSION-04 (small)' | Format-List Name, DriverName, PortName, PrinterStatus,
Comment
```

Expected: the new driver name, the *original* port, status *Normal*.

6.3 Evidence for the change record

- [ ] Before/after screenshots of the queue's Advanced tab, or the evidence JSON from Method B.
- [ ] Photograph of the configuration label showing 12 dots/mm.
- [ ] Photograph of the ruler label with a ruler alongside, showing 50 mm.
- [ ] Photograph of a LIMS-printed label next to the reference label, plus the scan result.
- [ ] Names of the requester and lab contact who accepted the result (recorded in the ticket, by role).

EXAMPLE · REFERENCE OUTCOME

In the reference engagement the ruler bar measured 50 mm, the LIMS reprint spanned the full label width with no text collisions, and the incident was resolved after the lab confirmed the labels.

6.4 Rollback

A driver swap is fully reversible because the old driver stays in the driver store. Choose the rollback that matches how you made the change:

Method used	Rollback
A · GUI	Queue **Properties › Advanced › Driver**: select the previous driver; re-apply the old printing defaults from your screenshots. For a wider problem, **Import printers from a file…** with the export taken in Step 1.
B · PowerShell	`Set-Printer -Name 'LAB-TORSION-04 (small)' -DriverName 'Zebra ZT620 (203 dpi)'`. The previous name is in the evidence JSON (*DriverBefore*).
C · PrintUI	`rundll32 printui.dll,PrintUIEntry /Sr /n "LAB-TORSION-04 (small)" /a "C:\Temp\CHG0012345-queue.dat"` restores the stored settings, including the driver name.
Whole server	`PrintBrm -R -F C:\Temp\server.printerExport`

WATCH OUT · ROLLBACK RESTORES THE DEFECT

Rolling back to the 203-dpi driver brings back undersized labels. Use rollback only if the new driver introduces a worse problem (for example jobs failing outright), and then open a problem record rather than leaving the lab on the wrong driver.

6.5 Failover-queue parity

Labs often have a second printer that takes over when the first is down, either through a redirect rule in BarTender or a fallback printer in the LIMS. If the failover queue has a different resolution, or the wrong driver, the very same defect reappears the moment failover happens, which is usually during an outage when nobody is looking at label size.

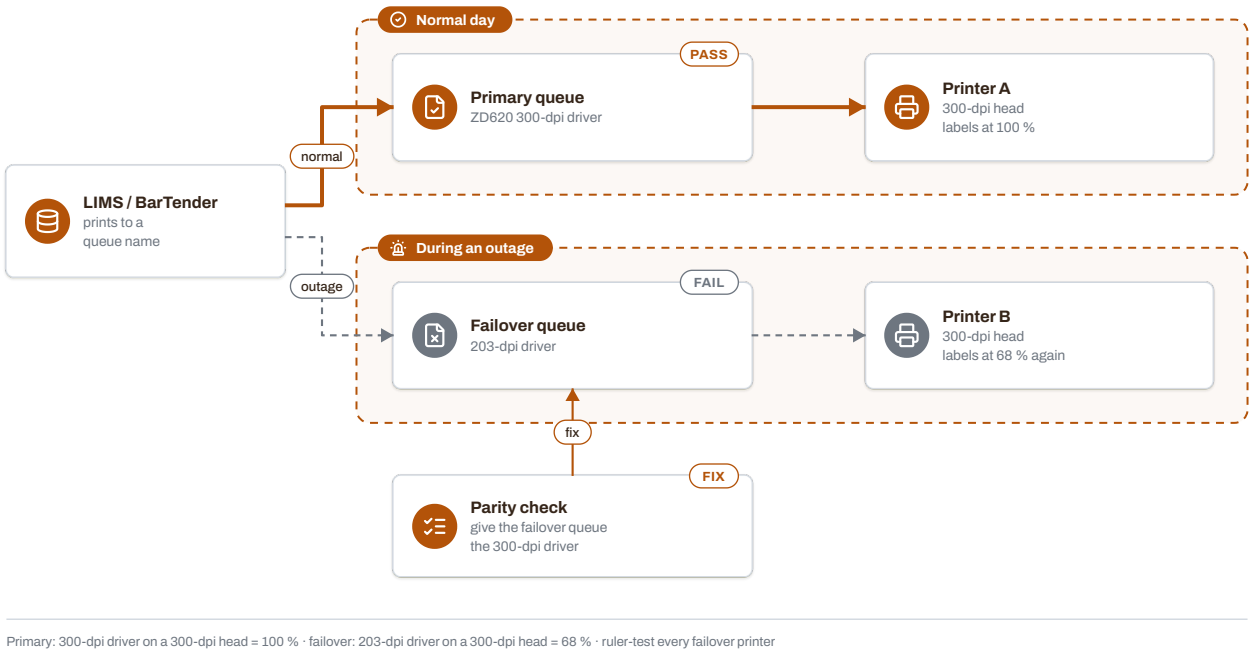


Figure 7. A failover queue is a second driver decision, so it must be ruler-tested against its own printer before an outage exposes it.

- [ ] List every queue that can receive this printer's jobs (redirects, LIMS fallbacks, pooled ports).
- [ ] Run the ruler test on each failover printer and confirm its queue driver matches *its* head.
- [ ] If the failover printer has a different resolution from the primary, either replace it with a matching unit or confirm that BarTender formats are resolution-independent (designed in physical units, rendered through the correct driver for each queue).

6.6 Prevention checklist

- [ ] When relocating or repointing a queue, verify the driver model *and* resolution match the physical unit before release; print a configuration label at handover.
- [ ] Record the head resolution (203/300/600) on the asset label, in the queue comment and in the CMDB.
- [ ] Use one driver family (Seagull) on BarTender servers; ZPL variants only.
- [ ] Run the 50 mm ruler test after any driver or hardware change. It costs one label and one command.
- [ ] Run the scheduled mismatch audit (Method D).

6.7 Troubleshooting

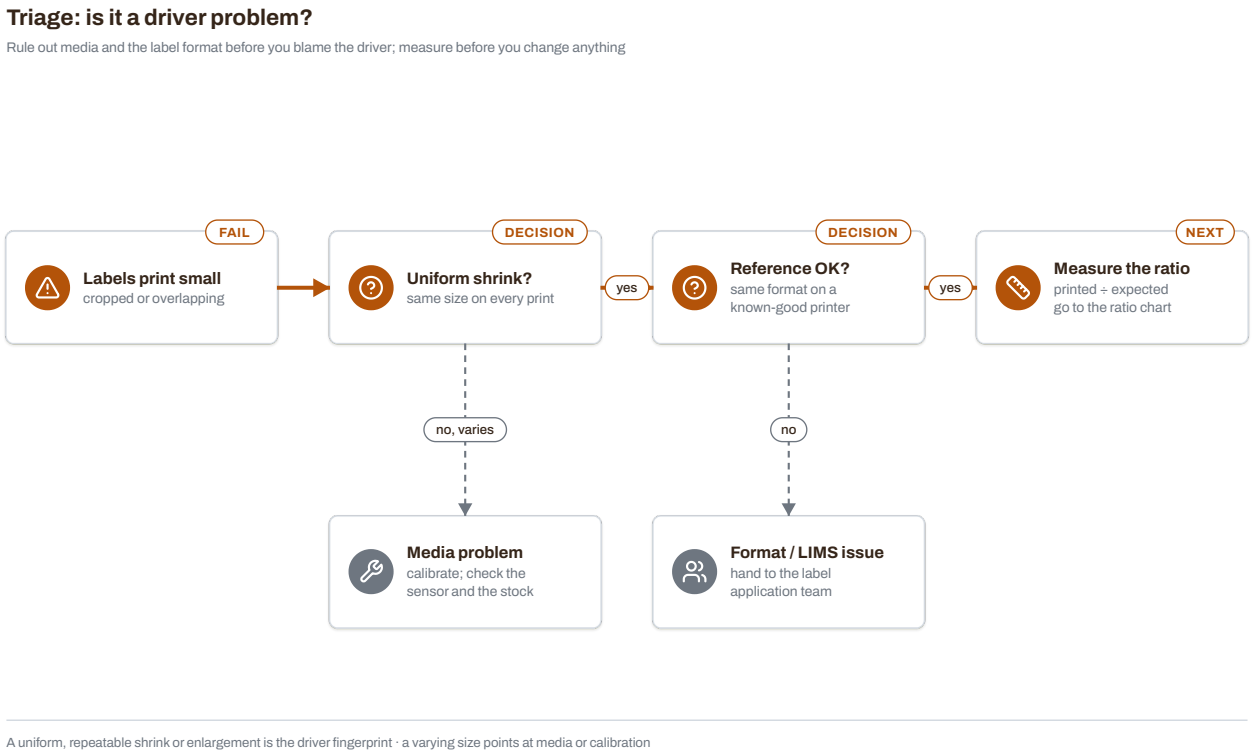


Figure 8. Rule out media and the label format first; only a uniform, repeatable size error justifies touching the driver.

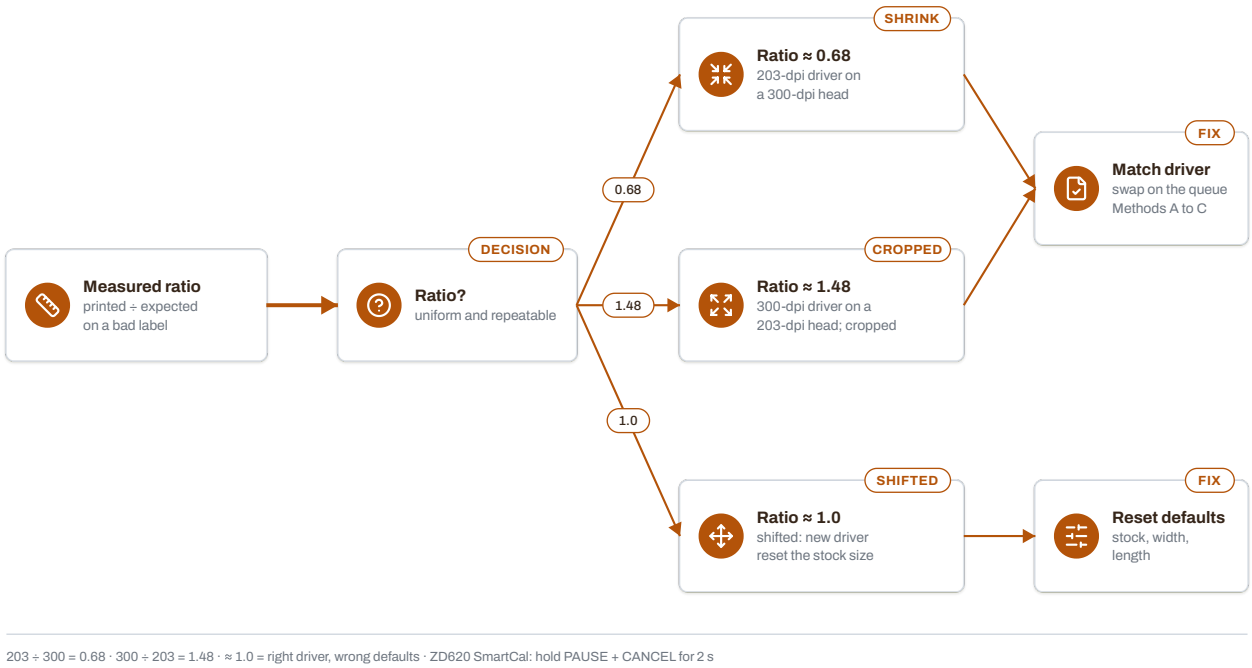


Figure 9. The ratio alone separates a too-low driver, a too-high driver and a correct driver with reset defaults.

Symptom	Likely cause and fix
Uniform shrink to ≈ 68 %	203-dpi driver on a 300-dpi head. Install and assign the 300-dpi ZPL driver.
Content too large, cropped right and bottom	300-dpi driver on a 203-dpi head (ratio ≈ 1.48). Assign the 203-dpi driver.
Correct size but shifted, or printing across two labels	Printing defaults reset by the new driver (stock size, orientation) or media not calibrated. Set stock in Printing Defaults; calibrate media (on a ZD620, hold PAUSE + CANCEL for two seconds for SmartCal).
Front-panel changes revert	Job commands (`^PW`, `^LL`, `^MD`, `^JUS`) override saved settings. Fix in the driver or BarTender format.
`Add-PrinterDriver`: "The specified driver does not exist in the driver store"	INF not staged, or the model name does not match the INF exactly. Stage with `pnputil /add-driver … /install`, then copy the model string from the INF.
Ruler test prints nothing	Port 9100 blocked, wrong IP, or printer paused. `Test-NetConnection 192.0.2.50 -Port 9100`; check the printer display.
Ruler label prints garbage text	Printer set to EPL or line-print mode. Print a configuration label and check the command language setting; restore ZPL.

CHAPTER 7

Summary, Glossary and Review

7.1 Chapter summary

- A 203-dpi driver on a 300-dpi head shrinks everything to 203 ÷ 300 = 68 %; the reverse enlarges and crops by 148 %. A uniform, repeatable ratio is the fingerprint.
- Swap the driver on the existing queue (GUI, PowerShell or PrintUI) so the name and port, and therefore the LIMS registration, are preserved. Back up first; reset printing defaults after.
- Prove the head with the raw 50 mm ZPL ruler test, prove the queue with an end-to-end LIMS reprint, and keep failover queues at parity.

7.2 Glossary

DPI	Dots per inch; printhead resolution (203 dpi = 8 dots/mm, 300 dpi = 12 dots/mm).
ZPL	Zebra Programming Language; the native command language of Zebra printers, framed by `^XA` … `^XZ`.
EPL	Eltron Programming Language; a legacy Zebra language kept for compatibility.
~HI	ZPL host identification query; the reply includes model, firmware and dots per millimetre.

Check your understanding

1. A label that should be 50 mm wide prints at 34 mm. Which resolution mismatch is most likely, and how did you work it out?
2. Why does changing the print width on the printer's front panel not fix the problem, and why does the change disappear?
3. Why is `Set-Printer -DriverName` preferred over deleting and re-creating the queue?
4. What does the 50 mm ruler test prove, and why must it be sent to port 9100 rather than printed through the queue?
5. After the fix, the lab's primary printer fails and jobs redirect to a second printer. What must you have checked to be sure labels stay correct?